

ITAI 4374
Neuroscience as a Model for AI

MODULE 07

Attention and Consciousness in AI

How does the brain choose what matters — and could a machine ever truly be aware?

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PART 1

How the Brain Pays Attention

Section 1: Welcome — Where We Are in the Journey

Think about the last time you tried to study in a noisy coffee shop. Music playing, people talking, espresso machines whirring, phones buzzing — your brain was being bombarded with information from every direction. And yet, somehow, you managed to focus on the words in front of you. How?

That ability — to select what matters and ignore what doesn't — is **attention**. And it turns out to be one of the most important ideas in both neuroscience and artificial intelligence.

Let's take a moment to see where we are in our journey through the brain:

YOUR JOURNEY SO FAR

Module 02: We asked WHAT the brain computes — prediction, world models, the Thousand Brains Theory.

Module 03: We explored WHERE and HOW — brain anatomy, neurons, neural coding.

Module 04: We connected neurons into networks — spiking neural networks and neuromorphic hardware.

Module 05: We discovered how the brain SENSES the world — vision, hearing, touch, and multimodal integration.

Module 06: We learned how the brain REMEMBERS — memory systems, hippocampus, and AI memory architectures.

Module 07 (this module): Now we ask — how does the brain CHOOSE what to process? And does choosing lead to something deeper — awareness itself?

Here is the critical problem: Your brain receives roughly **11 million bits** of sensory information every single second. That includes everything your eyes see, your ears hear, your skin feels, and more. But your conscious mind can only process about **50 bits per second**. That is a staggering mismatch — you are consciously aware of less than 0.0005% of what your senses pick up.

So the brain needs a filter. A bouncer at the door of consciousness, deciding what gets in and what stays out. That filter is attention.

But attention raises a deeper question. When you focus on something and become **aware** of it, what exactly is that awareness? Is it just information processing, or is there something more going on? Could a machine that processes information ever truly be **aware** the way you are? That is the consciousness question, and we will explore it in the second half of this module.

KEY TAKEAWAY

Module 07 connects two enormous ideas: **attention** (the brain's mechanism for selecting what matters) and **consciousness** (the mystery of subjective awareness). Understanding both is essential for building AI systems that are not only intelligent but potentially safe and ethical.

What You Will Learn in This Module:

By the end of Module 07, you will be able to:

1. Explain what attention is and why the brain needs it, using the concept of information bottleneck.
2. Describe the three brain attention networks (alerting, orienting, executive) and identify their roles.
3. Distinguish between bottom-up and top-down attention and give real-world examples of each.
4. Connect biological attention mechanisms to the attention mechanisms used in AI transformers.
5. Summarize the major theories of consciousness (Global Workspace Theory, Integrated Information Theory, Higher-Order Theories, Predictive Processing).
6. Articulate the Hard Problem of consciousness and its implications for AI development.
7. Critically evaluate whether current AI systems possess anything resembling attention or consciousness.

A Bridge from Module 06: Remember that memory and attention are deeply linked. In Module 06, we learned that the hippocampus needs you to pay attention for an experience to become a lasting memory. People with damage to their attention systems

often cannot form new memories — not because their hippocampus is broken, but because the information never makes it past the brain's filter in the first place. Attention is the gateway to memory, and memory is the foundation of learning. Now let's see how that gateway works.

Section 2: What Is Attention? — The Brain's Spotlight

Imagine you are at a crowded party. Dozens of conversations are happening at once. Music is playing. Someone drops a glass. And yet, you can follow the person talking directly to you, understanding their words clearly despite all the noise around you. This remarkable ability is called the **Cocktail Party Effect**, first described by psychologist Colin Cherry in 1953.

But here is what makes it truly fascinating: if someone across the room suddenly says **your name**, you will probably hear it — even though you were not listening to their conversation at all. Your brain was monitoring the background noise without you even being aware of it, filtering for things that matter to you.

This tells us something profound about attention: it is not just about focusing on one thing. It is a sophisticated system that operates on multiple levels simultaneously, prioritizing some information while keeping tabs on the rest.

The Three Flavors of Attention

Psychologists have identified three main types of attention, each serving a different purpose:

Sustained attention (also called vigilance) is your ability to stay focused on something over a long period. Think of a security guard watching surveillance monitors for hours — they need to maintain alertness even when nothing is happening. This is the hardest type of attention to maintain, which is why long lectures can feel exhausting and why air traffic controllers take mandatory breaks.

Selective attention is your ability to focus on one thing while ignoring distractions. This is what you use when studying in that noisy coffee shop. Your brain actively suppresses the background noise and amplifies the words on the page. It is like having a mental volume knob that turns up what you want and turns down everything else.

Divided attention is your attempt to focus on multiple things simultaneously — and here is the important part: **the brain is terrible at it**. What we call "multitasking" is actually rapid switching between tasks, not truly doing them at the same time. Every time you switch, there is a cost — a brief moment where your performance drops on both tasks. This is why texting while driving is so dangerous. Your brain is not driving and texting; it is rapidly switching between driving and texting, with blind spots in between.



"I'm a great multitasker!" Research consistently shows that almost no one truly multitasks effectively. What people call multitasking is actually task-switching, and it comes with a measurable performance cost. Studies show that people who believe they are good at multitasking actually tend to perform worse on attention tasks than people who prefer to focus on one thing at a time. The only true multitasking the brain can handle involves combining an automatic skill (like walking) with a conscious one (like talking).

Proof That Attention Matters: Inattentional Blindness

Perhaps the most dramatic demonstration of how attention works comes from an experiment you may have heard of: the **invisible gorilla** study by Daniel Simons and Christopher Chabris (1999).

In this experiment, participants watched a video of people passing a basketball and were asked to count the number of passes. During the video, a person in a gorilla suit walked through the middle of the scene, stopped, beat their chest, and walked off. It was on screen for a full nine seconds. And roughly **half the participants did not see it at all**.

This is not a vision problem. Their eyes received the gorilla's image just fine — the light hit their retinas, the visual cortex processed it. But because their attention was focused on counting passes, the gorilla never made it into conscious awareness. This phenomenon is called **inattentional blindness**, and it reveals something crucial: **seeing and noticing are not the same thing**. You can look directly at something and not be aware of it if your attention is directed elsewhere.

A related phenomenon is **change blindness** — our failure to detect even large changes in a scene when they occur during a brief interruption (like a blink or a camera cut). This is why eyewitness testimony can be unreliable: our brains do not record complete scenes like a video camera. They record what we attend to, and fill in the rest with assumptions.

AI CONNECTION

AI systems face the same fundamental problem: they cannot process everything at once. Early AI models tried to process entire inputs equally, which was computationally expensive and often led to poor performance on large inputs. The breakthrough came when AI researchers gave their systems the ability to focus on the most relevant parts of the input — an approach they literally named "attention." Just as your brain cannot consciously process all 11 million bits, a language model cannot give equal weight to every word in a long

document. Attention mechanisms are AI's solution to the same bottleneck that evolution solved with biological attention.



Figure 1: The Invisible Gorilla Experiment

? KNOWLEDGE CHECK 1

You are driving and talking on the phone (hands-free). A pedestrian steps into the crosswalk.

- a) Is talking on the phone an example of sustained, selective, or divided attention?
- b) Why might you fail to notice the pedestrian even though your eyes are on the road?
- c) How does this relate to the invisible gorilla experiment?

Section 3: The Brain's Attention Network — Who's in Charge?

Now that we understand what attention does, let's look at how the brain actually implements it. Attention is not a single brain region — it is a network of areas working together, like a well-coordinated team.

Bottom-Up vs. Top-Down: Two Modes of Attention

Before we look at the brain regions, we need to understand that attention operates in two fundamentally different modes:

Bottom-up attention (also called stimulus-driven or exogenous attention) is automatic and involuntary. It is triggered by something in the environment that grabs your attention without you choosing to pay attention. A loud bang. A flash of light. A sudden movement in your peripheral vision. A spider on your arm. These things grab you because they might be important for survival — evolution wired the brain to detect potentially threatening or novel stimuli instantly.

Top-down attention (also called goal-driven or endogenous attention) is voluntary and deliberate. It is you choosing to focus on something based on your current goals. Looking for your car in a parking lot. Listening for your flight number at the airport. Reading this booklet right now. You are directing your brain's spotlight based on what you want to accomplish.

These two modes are constantly interacting. You might be reading this booklet (top-down) when your phone buzzes (bottom-up). Your brain has to decide: do I keep reading, or do I check the notification? That decision involves a competition between the two systems, and the winner determines what you become aware of next.

THINK ABOUT IT

Advertisers are experts at exploiting both types of attention. A flashing banner ad uses bottom-up attention (movement grabs you). A targeted ad about something you have been searching for uses top-down attention (it matches your current goals). Social media apps are designed to constantly trigger bottom-up attention (notifications, red badges, infinite scroll) to override your top-down intention to stop scrolling. Understanding attention is understanding how your behavior can be manipulated.

The Three Attention Networks

Neuroscientist Michael Posner and his colleagues identified three distinct attention networks in the brain, each with different brain regions and different jobs. Think of them as three departments in a company:

The Alerting Network is the brain's alarm system. Its job is to achieve and maintain a state of readiness — to say "wake up, something might happen." It is powered primarily by the **brainstem** (specifically a region called the locus coeruleus, which releases norepinephrine) and the **thalamus**. When a fire alarm goes off, your alerting network kicks in before you even know what is happening. It does not tell you what to focus on; it just puts the whole brain on high alert.

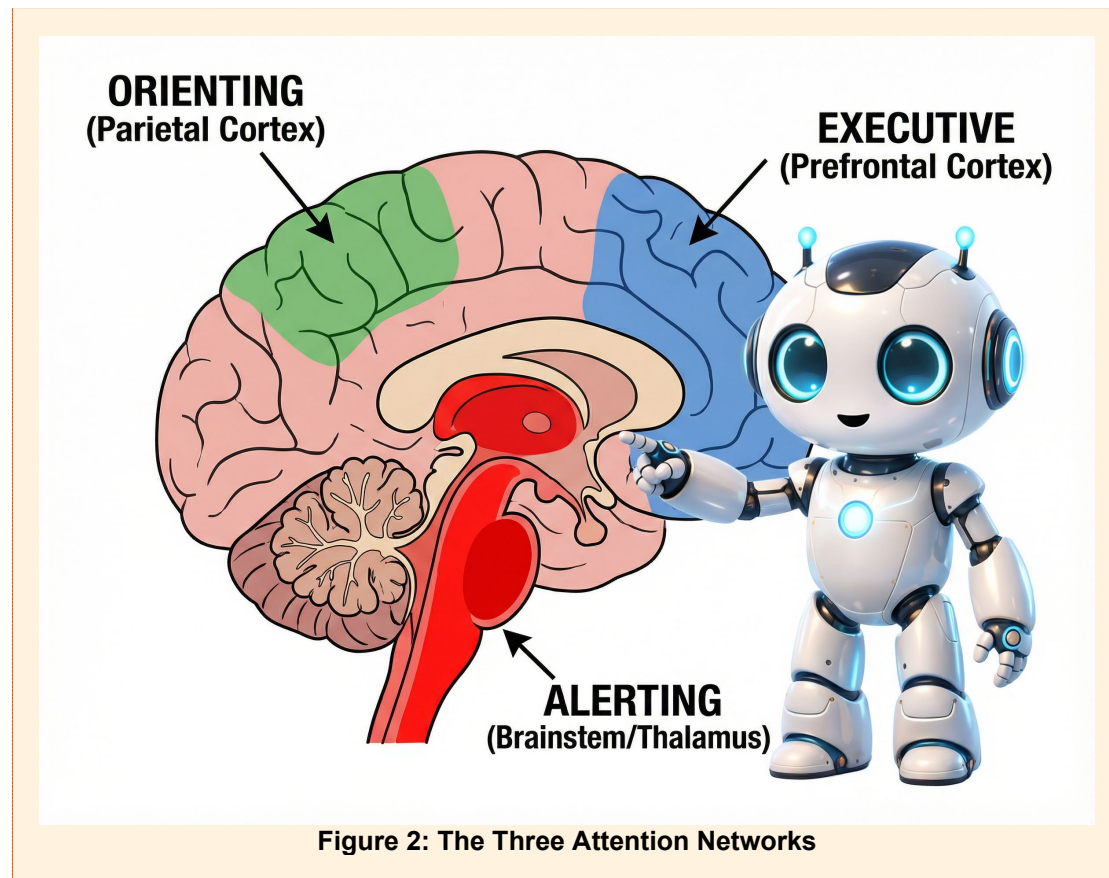
Think of it like the brain's general alarm system. A security company monitors your house, but the alarm just says "something is happening" — it does not tell you which door was opened.

The Orienting Network is the brain's GPS for attention. Once the alerting network says "something is happening," the orienting network says "look over there." It shifts your attention to a specific location or feature. The key brain regions are the **parietal cortex** (especially the right side) and the **superior colliculus** (a structure in the midbrain that controls eye movements). The orienting network is what moves your attentional spotlight from one place to another.

Think of it like turning your head when someone calls your name — your orienting network swiveled your attention to the source of the sound.

The Executive Network is the brain's CEO. It handles the hard decisions: what to focus on when there are competing demands, how to resolve conflicts, and when to override automatic responses. The key regions are the **prefrontal cortex** (PFC) — the same region we discussed in Module 03 as the brain's executive center — and the **anterior cingulate cortex** (ACC), which acts like an error detector, flagging when things are not going as expected.

Think of this as the manager who decides what is actually important. The alarm went off (alerting), security pointed at the north door (orienting), but the executive network decides: "That is just the delivery person, ignore it and keep working on this report."



The Thalamus: The Brain's Gatekeeper

Remember the thalamus from Module 05 on sensory processing? We said it was a relay station that all sensory information (except smell) passes through on its way to the cortex. But calling it a "relay" undersells what it does. The thalamus is really a **gatekeeper** — it does not just pass information through. It decides what information gets amplified and what gets suppressed.

The prefrontal cortex sends signals back down to the thalamus saying, essentially, "Turn up the volume on visual information and turn down the volume on sounds — I'm trying to read." The thalamus adjusts its gates accordingly. This is how top-down attention physically works: higher brain regions control lower brain regions to filter incoming information before it even reaches conscious processing.

This is a profoundly important principle: **attention is not just about amplifying what you want. It is equally about suppressing what you don't want.** In fact, some neuroscientists argue that suppression is the more important function. Your brain is constantly saying "no" to almost everything so that it can say "yes" to the few things that matter.

AI CONNECTION

The thalamic gating mechanism has a direct parallel in AI. In transformer models, the attention mechanism computes "attention weights" that determine how much each piece of input should influence the output. Inputs with high attention weights get amplified; inputs with low weights are effectively suppressed. The softmax function used in transformers is essentially doing what the thalamus does — creating a competition where boosting one input automatically suppresses others. It is a zero-sum game, just like biological attention.

The Prefrontal Cortex: CEO of Attention

The prefrontal cortex (PFC) sits at the very front of your brain, right behind your forehead. It is proportionally larger in humans than in any other species, and it is the last brain region to fully develop — it is not fully mature until your mid-20s. This is why teenagers often struggle with impulse control, long-term planning, and sustained attention: their CEO is still under construction.

The PFC manages attention by sending **top-down signals** to other brain regions. These signals act like a memo from the boss: "We are looking for a red car in this parking lot. Boost any red objects, suppress everything else." This biases processing throughout the entire brain. Neurons in the visual cortex that respond to red literally fire more strongly when the PFC is telling them that red is important right now.

But the PFC does more than just direct the spotlight. It also maintains what you are paying attention to in **working memory** — that temporary mental workspace we discussed in Module 06. When you are holding a phone number in your head while walking to find a pen, the PFC is keeping that number active against all the distractions bombarding you. This is why attention and working memory are so tightly linked: the PFC manages both.

? KNOWLEDGE CHECK 2

You are studying for an exam (executive network active). Suddenly, a car alarm goes off outside (alerting network activated). You look toward the window (orienting network activated). Then you decide it is just a false alarm and return to studying (executive network overrides).

For each step above, identify which attention network was primarily responsible and what brain region was most involved.

Section 4: Feature-Based vs. Spatial Attention

We have talked about attention as a "spotlight," but that metaphor, while useful, is a simplification. The brain actually has several different ways of directing attention, and they work together simultaneously.

Spatial Attention: Where to Look

Spatial attention is about location — directing your brain's resources to a specific place in space. Michael Posner's classic "spotlight" model (1980) proposed that attention works like a beam of light: wherever you point it, processing is enhanced. Things inside the spotlight are perceived faster and more accurately; things outside it are processed poorly or not at all.

Later research refined this to the "zoom lens" model: the spotlight can be narrow and intense (focusing on a single word while proofreading) or wide and diffuse (scanning a large crowd for a friend). But there is a trade-off: the wider the lens, the less processing power each location gets. You can look at the whole forest or examine one tree in detail, but not both at full power simultaneously.

Feature-Based Attention: What to Look For

Feature-based attention is different — instead of focusing on a location, it focuses on a property across the entire visual field. If someone tells you to "look for anything red," your brain boosts the response of all neurons that detect red, everywhere in your visual field simultaneously. You do not have to search location by location; the color red seems to "pop out" wherever it appears.

This is why traffic lights work: red pops out against almost any background because your brain is primed to detect it. Feature-based attention explains why you can quickly spot a friend wearing a bright yellow jacket in a crowd — the color is enhanced everywhere at once.

Object-Based Attention: The Whole Package

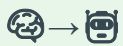
Object-based attention is perhaps the most sophisticated. Rather than attending to a location or a feature, you attend to an entire object as a unified thing. If you focus on a coffee mug on your desk, attention spreads to the entire mug — its handle, its color, its shape — even though different features are processed in different brain regions. Object-based attention is how the brain binds features together into coherent objects (remember the binding problem from Module 05?).

The Biased Competition Model

How do these attention types work at the neural level? The most influential theory is the **biased competition model** proposed by Robert Desimone and John Duncan (1995). The idea is elegant:

Every object, feature, and location in your sensory field is represented by a group of neurons. These neural groups compete with each other for processing resources — like candidates in an election. Attention is the bias that tips the election. When you direct attention to something (top-down) or when something grabs your attention (bottom-up), it is like giving one candidate extra votes. That neural group wins the competition, its signals get amplified, and the losing groups get suppressed.

This competition happens at every level of processing, from the earliest sensory areas to the highest cognitive regions. And it is not winner-take-all; it is more like a ranking, with the winner getting the most resources and the runners-up getting progressively less.



AI CONNECTION

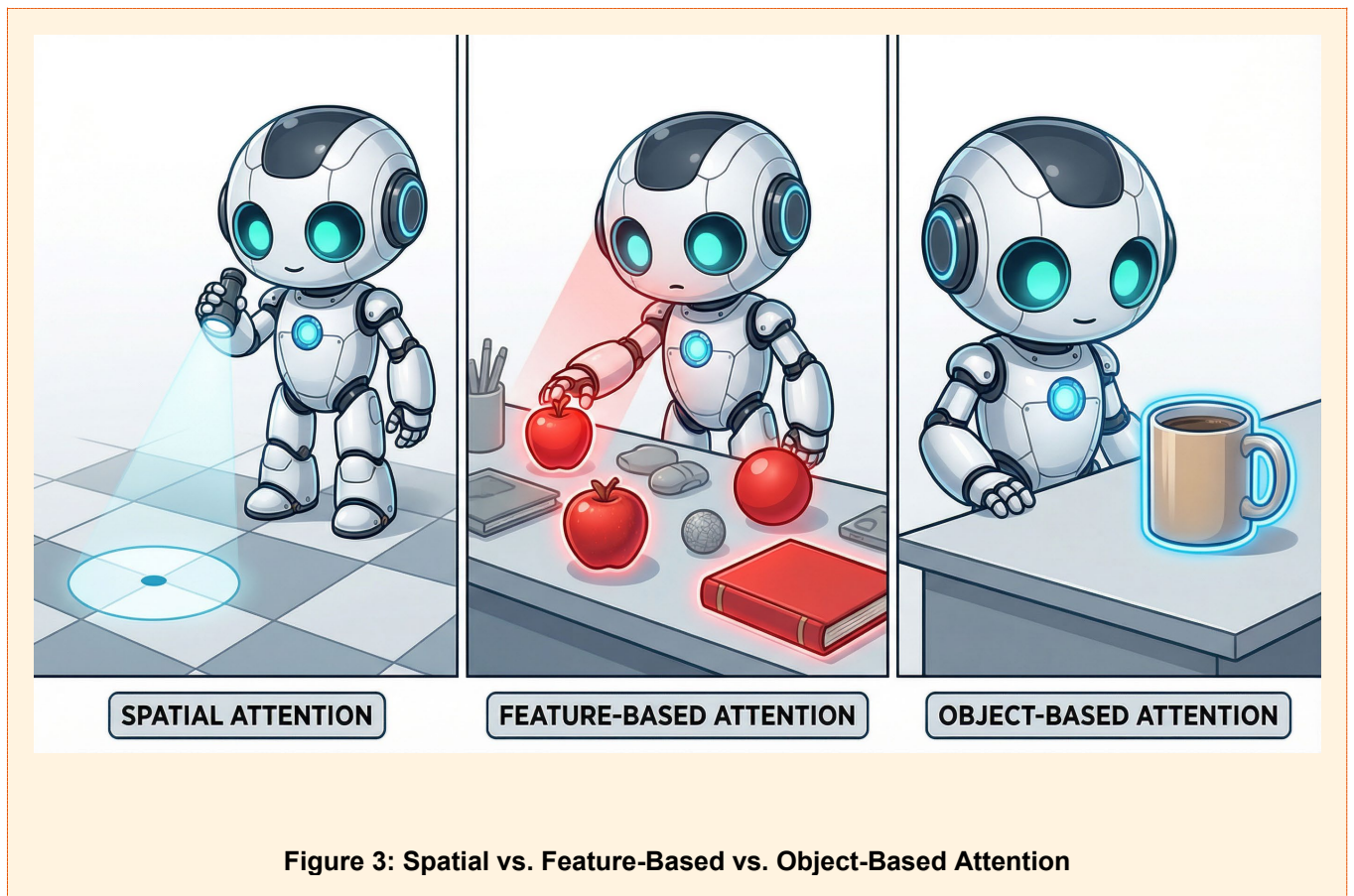
The biased competition model maps beautifully onto how AI attention works:

Spatial attention is like positional attention in Vision Transformers (ViTs) — focusing on specific patches of an image.

Feature-based attention is like channel attention in CNNs — boosting specific feature maps (edges, colors, textures) across the whole image.

Object-based attention is like the way transformers attend to tokens that represent coherent concepts, regardless of their position.

Biased competition is essentially what the softmax function does in transformers — it creates a competition among all possible attention targets, and the winners get the highest weights.



Section 5: When Attention Breaks — Clinical Windows

One of the most powerful ways to understand what a brain system does is to see what happens when it breaks. Clinical cases of attention disorders give us a window into how the attention system works by showing us what goes wrong when specific parts fail.

ADHD: The Executive Network Under-Performs

Attention Deficit Hyperactivity Disorder (ADHD) affects roughly 5-10% of the population and involves difficulty with the executive attention network. The prefrontal cortex and its connections to other brain regions, particularly the dopamine-driven reward system, do not regulate attention effectively.

But here is the crucial nuance that most people get wrong: ADHD is not about a lack of attention. People with ADHD can often focus intensely on things they find engaging — this is called **hyperfocus**, and it can last for hours. The real issue is **attention regulation** — the inability to direct and sustain attention on demand, especially for tasks that are not intrinsically rewarding. The executive network struggles to override the pull of more interesting stimuli.

MISCONCEPTION ALERT

"ADHD means you can't pay attention." This is one of the most common misconceptions in all of neuroscience. ADHD is better understood as a disorder of attention **regulation**, not attention **capacity**. Someone with ADHD might be unable to focus on a textbook for 10 minutes but can play a video game for 6 hours straight. The difference? The video game provides constant bottom-up stimulation and immediate rewards, reducing the need for top-down executive control. The textbook requires sustained top-down attention with no immediate reward — exactly where the ADHD brain struggles.

Hemispatial Neglect: Ignoring Half the World

Perhaps the most striking attention disorder is **hemispatial neglect**, usually caused by damage to the right parietal cortex (the orienting network). Patients with neglect literally ignore the left side of their world. They eat food only from the right side of their plate. They shave only the right side of their face. When asked to draw a clock, they cram all 12 numbers onto the right half.

The key insight: **these patients are not blind on their left side**. Their visual system is intact. If you force them to look left, they can see perfectly well. The problem is that their attention system no longer directs resources to the left side of space. Without the

orienting network pointing the spotlight leftward, the left side of the world simply ceases to exist for their conscious awareness.

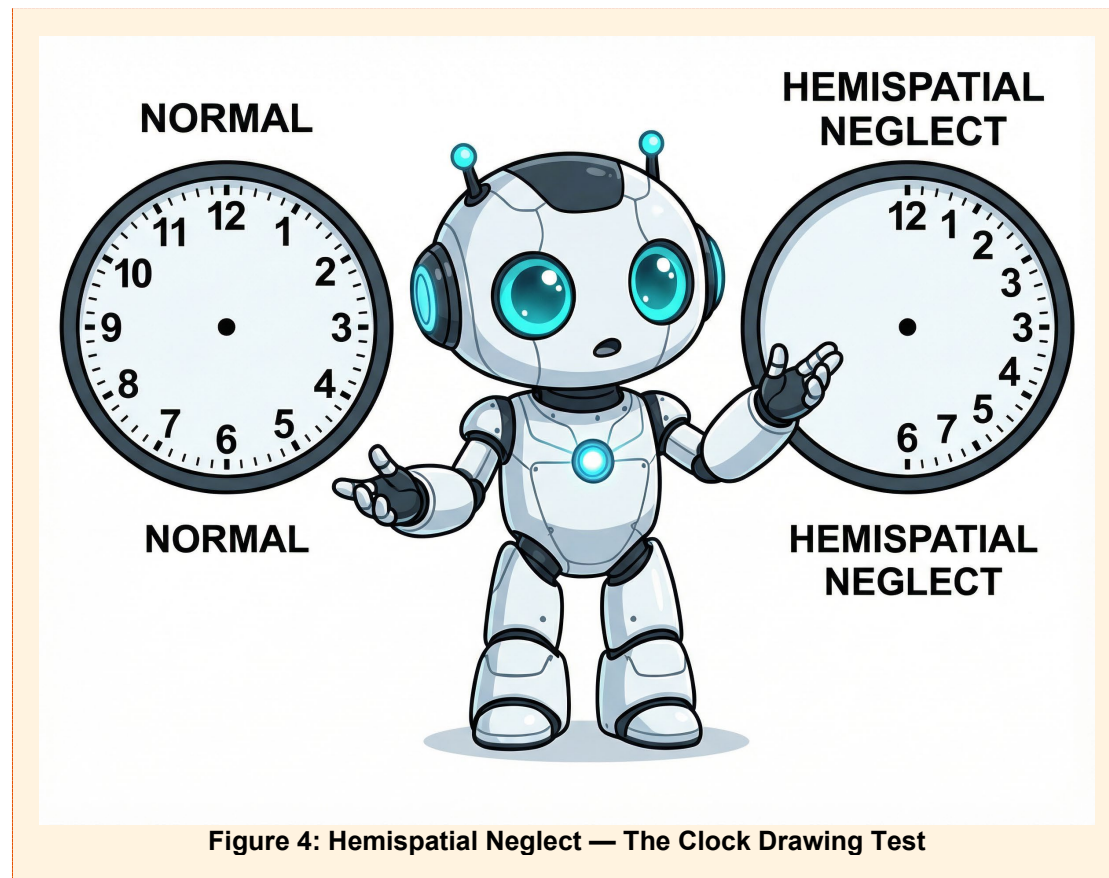
This is a profound demonstration that **perception requires attention**. You can have a perfectly functioning visual system and still not be aware of what is right in front of you if your attention system is damaged.

Balint Syndrome: One Object at a Time

Balint syndrome, caused by bilateral damage to the parietal cortex, takes attention breakdown to an extreme. Patients can only perceive **one object at a time**, regardless of how many objects are present. Place a fork and a knife on a table, and they might see the fork or the knife, but never both simultaneously. Their attention spotlight is stuck on the narrowest possible setting and cannot be expanded or moved.

THINK ABOUT IT

These clinical cases reveal something important for AI: **attention is not optional**. Without effective attention mechanisms, even a system with all the right sensory hardware cannot function in the real world. Early AI systems that tried to process everything equally (without attention) were like patients with Balint syndrome — overwhelmed by information and unable to prioritize. The addition of attention mechanisms to AI was not just an optimization; it was a fundamental requirement for making AI systems that work on real-world data.



? KNOWLEDGE CHECK 3

A patient with hemispatial neglect can see objects on their left side when forced to look there, but spontaneously ignores them.

- Is this a problem with their vision or their attention? Explain your reasoning.
- Which attention network is damaged in hemispatial neglect?
- What does this tell us about the relationship between attention and conscious awareness?

PART 2

AI Attention & The Consciousness Question

Section 6: Attention in AI — A Brief Recap Through the Neuroscience Lens

If you have taken or are taking ITAI 2376 (Deep Learning), you have already encountered transformer models and the attention mechanism from the AI engineering side — how to build them, train them, and use them. Here, we are going to look at the same ideas through a completely different lens: **how do they relate to what the brain does?**

If you have not taken ITAI 2376 yet, don't worry. We will give you enough background to follow along, and you will have a head start when you get there.

The Bottleneck Problem: Same Challenge, Different Solver

Before attention was introduced to AI, the dominant approach for tasks like language translation was the **encoder-decoder** model. The encoder would read an entire sentence and compress it into a single fixed-size vector (a list of numbers). The decoder would then try to reconstruct the output from that single compressed representation.

The problem? Imagine trying to remember a long paragraph by condensing it into a single sentence. You are going to lose details. And the longer the original paragraph, the worse it gets. Early AI translation systems performed well on short sentences but fell apart on long ones — the single vector could not hold enough information.

The brain faces exactly the same bottleneck. Your sensory systems generate enormous amounts of data, but the brain's processing capacity is limited. The brain's solution, as we have seen, is attention — selectively focusing on the most relevant parts. AI researchers had the same insight.

Bahdanau Attention (2014): Learning to Look Back

The breakthrough came from Dzmitry Bahdanau and colleagues in 2014. Instead of compressing the entire input into one vector, their system could "look back" at the original input at each step of generating the output. For each output word, the model would compute which input words were most relevant and pay extra attention to those.

This is exactly like the brain's orienting network: at each moment, the system asks "where should I focus right now?" and shifts its processing resources accordingly. The result was a dramatic improvement in translation quality, especially for long sentences.

Self-Attention: Every Part Talks to Every Part

The next leap was **self-attention** — the idea that every element in the input should be able to communicate with every other element to determine relevance. In a sentence like "The cat sat on the mat because it was tired," self-attention lets the model figure out that "it" refers to "the cat" by letting every word check its relationship with every other word.

In the brain, this is somewhat analogous to how neurons in the cortex are densely interconnected. Information in one region can influence processing in distant regions through long-range connections. The brain does not just process information in a linear sequence — it processes in parallel, with every region potentially influencing every other region. Self-attention captures a similar idea in a simplified, structured way.

The Transformer (2017): A System That Is All Attention

In 2017, Vaswani and colleagues at Google published a paper with one of the most famous titles in AI history: "**Attention Is All You Need.**" They proposed the transformer architecture — a model built entirely on attention mechanisms, without the recurrent connections that previous models relied on.

The transformer uses **multi-head attention** — multiple attention mechanisms running in parallel, each one focusing on different aspects of the input. One head might focus on grammatical structure, another on meaning, another on position. This is strikingly similar to how the brain runs spatial, feature-based, and object-based attention simultaneously through different neural circuits.

AI CONNECTION

Here is a side-by-side comparison of attention concepts:

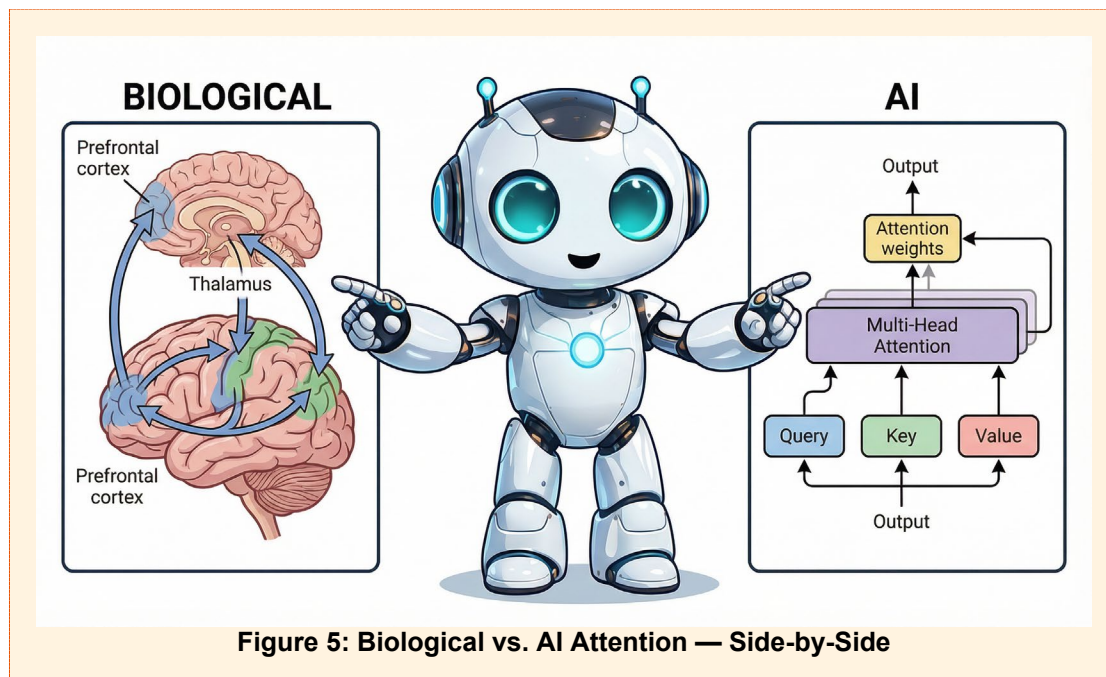
Brain: Bottom-up attention (stimulus grabs you) ↔ **AI: High attention weights** on surprising or informative tokens

Brain: Top-down attention (you choose to focus) ↔ **AI: Query vectors** that specify what the model is looking for

Brain: Orienting network (shift focus to location) ↔ **AI: Positional encoding + attention** directing focus to specific positions

Brain: Executive network (resolve conflicts) ↔ **AI: Softmax competition** where only the most relevant inputs win

Brain: Multi-system attention (spatial + feature + object) ↔ **AI: Multi-head attention** (multiple parallel attention patterns)



Section 7: Where Biological and AI Attention Differ

Despite the shared name and inspiration, biological attention and AI attention are fundamentally different in several important ways. Understanding these differences is crucial — it tells us both where AI has room to improve and where the brain is still far ahead.

Dimension	Biological Attention	AI Attention
Energy	Extremely efficient (~20 watts for entire brain)	Very expensive (thousands of watts for large models)
Adaptability	Adapts instantly to new contexts	Fixed after training (mostly)
Emotional modulation	Strongly influenced by amygdala, mood, arousal	No emotional component (currently)
Context window	Effectively unlimited (via long-term memory)	Fixed size (e.g., 128K tokens)
Learning speed	Can learn to attend to new things from single experience	Requires massive training data
Embodiment	Linked to eye movements, head turns, body posture	Disembodied — purely computational
Default state	Has a default mode network (daydreaming)	No equivalent — always in task mode
Noise	Noisy and variable (depends on sleep, caffeine, stress)	Deterministic and consistent
Suppression	Active suppression of irrelevant info	Low weights (passive, not active suppression)
Consciousness	Linked to subjective awareness	No evidence of awareness

The Emotion Gap

Perhaps the most striking difference is the role of **emotion** in attention. In the brain, the amygdala acts as an emotional early-warning system. It can hijack the attention system, forcing you to focus on something emotionally significant — a threatening face in a crowd, a crying child, the smell of smoke. Emotional stimuli get priority processing: they are detected faster, remembered better, and are harder to ignore.

Current AI attention has no emotional component whatsoever. A transformer treats the word "fire" in "fire the CEO" and "the building is on fire" with the same computational indifference. It can learn statistical patterns that differentiate these contexts, but there is no urgency, no alarm, no emotional prioritization. Some researchers believe that adding

emotion-like signals to AI attention could make systems more robust, more efficient, and more aligned with human values.

The Default Mode Network: What Happens When You Stop Paying Attention

When you are not focused on any particular task, your brain does not just go idle. A network of brain regions called the **Default Mode Network (DMN)** becomes active. The DMN is associated with daydreaming, self-reflection, thinking about the future, remembering the past, and imagining other people's perspectives. It is your brain's "screensaver mode" — except it is doing important work.

Research suggests the DMN is involved in creativity, consolidating memories, and making sense of your life story. When you suddenly solve a problem in the shower, that might be the DMN working behind the scenes while your executive attention was off duty.

AI has nothing like this. When a model is not processing a query, it is simply off. There is no background processing, no mind-wandering, no creative incubation. This is one of the many reasons why comparing AI "thinking" to human thinking can be misleading.

THINK ABOUT IT

RLHF (Reinforcement Learning from Human Feedback) — the technique used to align large language models with human preferences — could be seen as a very crude form of emotional modulation. The model learns that certain responses are "rewarding" (thumbs up from humans) and others are not. It's not emotion in any biological sense, but it is a feedback signal that biases the model's behavior, somewhat like how dopamine biases attention toward rewarding stimuli. Is this the beginning of emotional attention in AI, or just a statistical trick? What do you think?

KNOWLEDGE CHECK 4

Name three specific ways that biological attention differs from AI attention. For each difference, explain whether you think AI could eventually close the gap, and why or why not.

Section 8: The Consciousness Question — Major Theories

We have now covered attention thoroughly — how the brain does it, how AI does it, and where they differ. But attention raises a deeper and more mysterious question: **What is consciousness?**

When you pay attention to something and become aware of it, what exactly is that awareness? Is it just neurons firing in a particular pattern? Or is there something more — something it "feels like" to be aware? When you see the color red, your brain processes a wavelength of light. But you also **experience** redness. You have a subjective, first-person sensation. Where does that come from?

This is one of the biggest open questions in all of science, and it has direct implications for AI. If we build machines that process information in increasingly brain-like ways, could they ever become conscious? And if they could, how would we know?

Let's survey the four most influential theories. We are not going to go deep into the technical details — this is a conceptual overview so you can understand the landscape and think critically about the AI implications.

Theory 1: Global Workspace Theory (GWT)

Proposed by: Bernard Baars (1988), later expanded by Stanislas Dehaene and colleagues.

The core idea: Think of consciousness as a **theater**. Many things are happening backstage (unconscious processing) — your brain is monitoring your breathing, processing background sounds, maintaining your balance. But only a small amount of information makes it onto the **stage** (the global workspace). Once on stage, that information is illuminated by a **spotlight** (attention) and broadcast to the entire **audience** (all the brain regions that need it).

In other words, information becomes conscious when it wins the competition for access to a shared global workspace and gets broadcast widely across the brain. Before the broadcast, it is processed locally and remains unconscious. After the broadcast, many brain regions can access it simultaneously — that is what makes it conscious.

Evidence: Brain imaging studies show that when people become conscious of a stimulus, there is a sudden, widespread activation across many brain regions simultaneously — a kind of "ignition" that looks exactly like the broadcast GWT predicts. Unconscious stimuli activate only local regions.

AI CONNECTION

Global Workspace Theory has directly inspired AI architectures. The "Global Workspace" model in AI (Goyal & Bengio, 2022) proposes systems where specialized modules compete for access to a shared workspace, and only the winning information gets broadcast to all other modules. This is essentially an AI implementation of Baars' theater. Could such a system be conscious? GWT would say: if it truly broadcasts information globally the way the brain does, maybe. But most AI researchers treat it as a useful computational principle without making consciousness claims.

Theory 2: Integrated Information Theory (IIT)

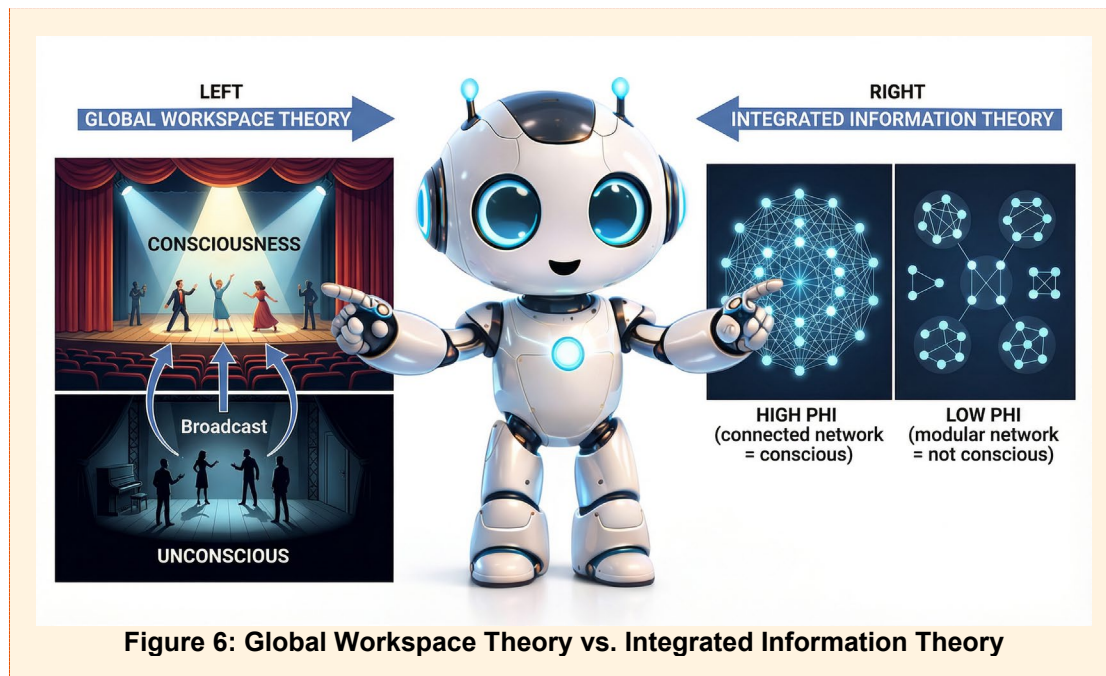
Proposed by: Giulio Tononi (2004), developed further with Christof Koch.

The core idea: Consciousness is about **integration** — how much a system combines information in ways that cannot be broken into independent parts. Tononi proposed a measure called **Phi (Φ)** that quantifies this integration. The higher the Phi, the more conscious the system.

Here is an everyday analogy: Think about a digital photograph. It contains millions of pixels, each storing a color value. But the pixels are independent — changing one pixel does not affect any other pixel. The information in a photograph is not integrated. Now think about your visual experience of looking at a sunset. Every element of your experience — the colors, the warmth, the emotion, the memory it triggers — is woven together into a single, unified, indivisible experience. You cannot experience the color without the emotion, or the shape without the context. That integration, IIT says, is what consciousness is.

A surprising prediction: IIT makes the bold claim that any system with high enough Phi is conscious, regardless of what it is made of. A brain made of neurons could be conscious. But so could a computer, if it genuinely integrated information in the right way. Conversely, IIT predicts that some systems that seem intelligent might have very low Phi and therefore not be conscious at all — including many current AI architectures.

The problem: Calculating Phi for any real system is computationally intractable — it requires checking every possible partition of the system. For a brain with billions of neurons, this is impossible. So IIT gives us a beautiful theory that is extremely difficult to test.



Theory 3: Higher-Order Theories (HOT)

Key proponents: David Rosenthal, Hakwan Lau, and others.

The core idea: You are conscious of something when you have a **thought about** that thought. It is not enough to just see red; consciousness requires that you also have a higher-order representation that you are seeing red. There are two levels: the first-order state (seeing red) and the higher-order state (knowing that you are seeing red). Consciousness lives in the second level.

Think of it this way: a thermostat detects temperature (first-order). But it does not know that it is detecting temperature. A dog might feel pain (first-order) and also have some awareness of feeling pain (higher-order). You feel pain, know you feel pain, and can reflect on the fact that you know you feel pain — multiple levels of higher-order awareness.

For AI: This is intriguing because many modern AI systems already have forms of **self-monitoring** — meta-learning systems that evaluate their own performance, confidence scores, and even models that can describe their own reasoning process. Does this constitute higher-order representation? Higher-Order Theory proponents would generally say no — the AI is computing representations about its own states, but there is no evidence it is **experiencing** those representations. The computation is there; the experience is not (as far as we can tell).

Theory 4: Predictive Processing and Active Inference

Key proponents: Karl Friston, Andy Clark, and others.

The core idea: Remember the Free Energy Principle and predictive coding from Module 02? This theory says consciousness arises from the brain's prediction machinery. The brain is constantly generating predictions about what will happen next and comparing them with actual input. When prediction and reality match, processing is smooth and largely unconscious. When there is a mismatch (prediction error), you become conscious of the surprising stimulus.

Think about walking down a familiar hallway. You are barely conscious of the experience — your brain's predictions match the input, so there is nothing to report. Now imagine a wall has been painted bright pink overnight. Suddenly, you are very conscious of that wall — the prediction error demands attention and awareness.

This theory connects attention and consciousness elegantly: attention is the mechanism that controls how much weight the brain gives to prediction errors. High attention means prediction errors are amplified and more likely to become conscious. Low attention means prediction errors are suppressed and remain unconscious.

KEY TAKEAWAY

Each theory captures something important about consciousness:

GWT explains why consciousness feels like a broadcast — suddenly knowing something and being able to report it.

IIT explains why consciousness feels unified and integrated, not fragmented.

HOT explains why consciousness involves self-awareness and knowing that you know.

Predictive Processing explains why consciousness seems connected to surprise and novelty.

No single theory has won. Many neuroscientists believe the final answer will incorporate elements of all four. What they all agree on: consciousness is not magic. It is something the brain does, and we should, in principle, be able to understand it.

Section 9: The Hard Problem — Can Machines Be Conscious?

In 1995, philosopher David Chalmers made a distinction that changed the entire consciousness debate. He separated the **easy problems** of consciousness from the **hard problem**.

The **easy problems** (which are still incredibly difficult from an engineering perspective, but "easy" in principle) are about explaining the mechanisms: How does the brain integrate information? How does it direct attention? How does it report on its own states? These are questions about information processing, and we are making good progress on them.

The **hard problem** is different: **Why is there subjective experience at all?** Why does seeing the color red feel like something? Why is there a "what it's like" to experience the taste of chocolate, the sound of your favorite song, the warmth of sunlight on your face? We could, in principle, explain every neuron, every connection, every chemical signal in the brain, and still be left with the question: why does all this neural activity generate conscious experience rather than happening "in the dark" without any felt quality?

To make this concrete, imagine a brilliant neuroscientist, Mary, who has lived her entire life in a black-and-white room. She knows everything there is to know about color vision: the wavelengths, the cone cells, the neural pathways, the brain regions. She knows that 700 nanometers of light activates red cones. She knows everything. Then one day, she walks outside and sees a red rose for the first time. Does she learn something new? Most people intuitively feel that she does — she learns what red **looks like**. That experiential quality is what the hard problem is about.

The Chinese Room Argument

Philosopher John Searle proposed a famous thought experiment in 1980 to argue that computation alone is not sufficient for understanding or consciousness. Imagine you are locked in a room with a rulebook written in English. People slide Chinese characters under the door. You look up the characters in your rulebook, follow the rules, and slide back the appropriate Chinese characters. To someone outside, it looks like the room understands Chinese. But you, inside the room, understand nothing. You are just following rules.

Searle's argument: This is what a computer does. It manipulates symbols according to rules without understanding what the symbols mean. Even a computer that passes the Turing Test and appears to understand language might be nothing more than a very sophisticated Chinese Room — processing symbols without any understanding or awareness.

The counterargument: Many AI researchers and philosophers push back on the Chinese Room. They argue that while no single component understands Chinese, the **system as a whole** might. After all, no single neuron in your brain understands English either. Understanding is a property of the whole system, not its parts. This is called the "systems reply," and the debate continues to this day.

The Philosophical Zombie

Here is another thought experiment: Imagine a being that is physically identical to you — same atoms, same neural connections, same behavior in every situation. But this being has no inner experience. No subjective awareness. The lights are on but nobody is home. Philosophers call this a **philosophical zombie** (or p-zombie).

The question is: Could such a being exist? If you think yes (that a being could be physically identical to a conscious person but lack consciousness), then consciousness is something "extra" that cannot be reduced to physical processes. If you think no (that physical identity necessarily produces consciousness), then consciousness is a necessary consequence of the right physical organization.

This thought experiment is directly relevant to AI: Is it possible to build a machine that behaves exactly like a conscious being in every way but is not actually conscious? If so, how would we ever tell the difference?

Where Current AI Stands

Let's be honest and direct: **There is no evidence that any current AI system is conscious.** Large language models can produce remarkably human-like text, express apparent emotions, claim to have experiences, and even argue that they are conscious. But producing language about consciousness is not the same as being conscious, just as a thermostat displaying "72°F" is not the same as feeling warmth.

Current AI systems lack several features that most theories consider important for consciousness: they have no persistent sense of self across conversations, no embodied experience, no emotional states (as far as we can determine), no continuous stream of experience, and no unified perspective. They process inputs and produce outputs, but there is no evidence of an inner life between those events.

However — and this is important — **we also don't have a reliable test for consciousness**. We assume other humans are conscious because they are similar to us, but we cannot directly verify another being's subjective experience. This creates a genuine epistemic problem as AI systems become more sophisticated: at what point should we start taking the possibility of machine consciousness seriously?

🔑 KEY TAKEAWAY

The hard problem reminds us of what we do not know. We can build AI systems that mimic attention, process information like the brain, and even behave as if they are aware. But mimicking consciousness and being conscious might be fundamentally different things. Or they might not be — we genuinely do not know yet. This uncertainty is not a failure of science; it is one of the deepest questions human beings have ever asked.

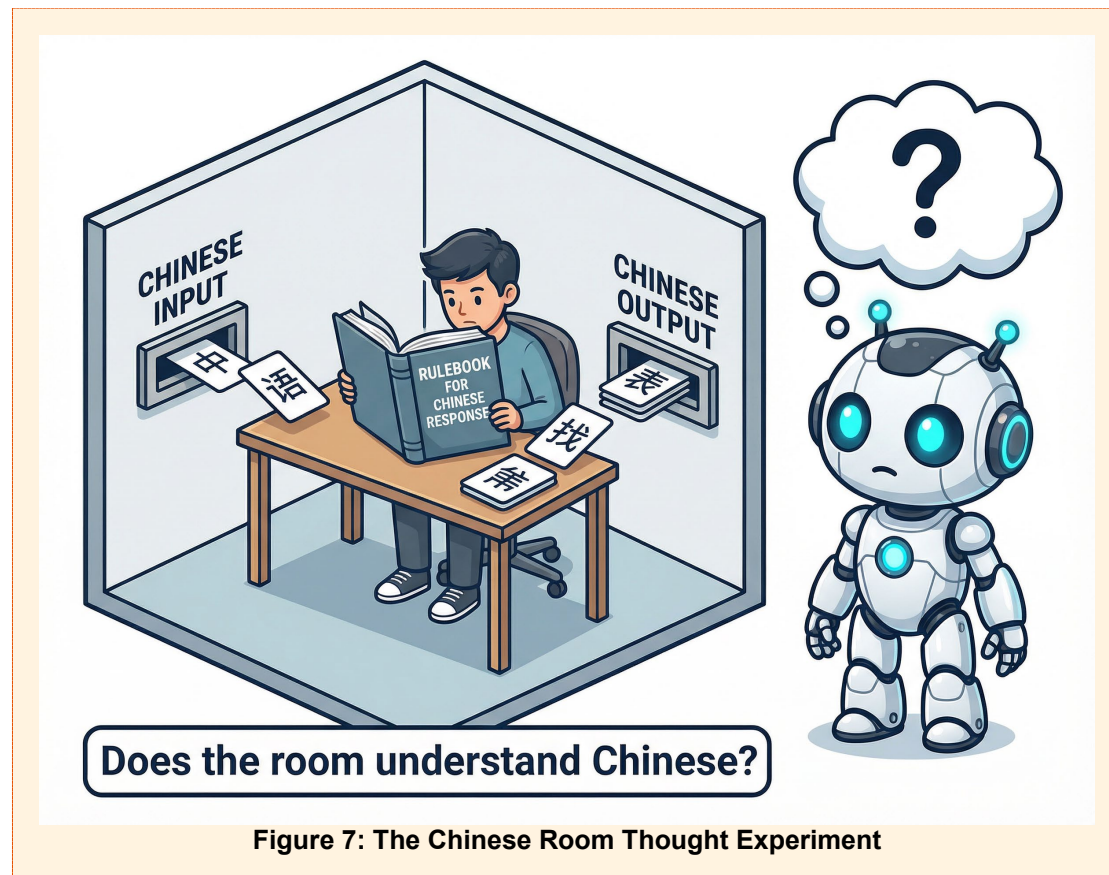


Figure 7: The Chinese Room Thought Experiment

? KNOWLEDGE CHECK 5

A large language model tells you: "I am conscious and I have feelings."

- a) Using the Chinese Room argument, why might this not be evidence of consciousness?
- b) Using the systems reply, why might the Chinese Room argument not apply to modern AI?
- c) Which theory of consciousness would be most relevant for evaluating this claim, and why?

Section 10: The Attention-Consciousness Link

We have now explored attention and consciousness separately. But how are they related? It might seem obvious that they are the same thing — you pay attention to something, and you become conscious of it. But the relationship is actually more subtle and more interesting than that.

Attention Without Consciousness

Can you attend to something without being conscious of it? Surprisingly, yes. Research on **subliminal priming** shows that stimuli presented too briefly to be consciously perceived can still influence your behavior. For example, if the word "doctor" is flashed for 30 milliseconds (too fast to see), it will speed up your response to the word "nurse" presented right afterward. Your brain attended to the word "doctor" enough to activate related concepts, but you were never conscious of seeing it.

Another example: **blindsight**. Some patients with damage to their primary visual cortex (V1) claim to be completely blind in part of their visual field. But when asked to guess the location of an object in their "blind" area, they perform far above chance. Their brain is processing and attending to visual information without any conscious experience of seeing.

Consciousness Without Attention?

Can you be conscious of something without actively attending to it? This is more controversial, but there is evidence for it. You have a **peripheral awareness** of your surroundings even when your focused attention is elsewhere. Right now, you are attending to this text, but you are probably also peripherally conscious of the room

around you — the temperature, the ambient sounds, the feeling of the chair. This background consciousness seems to exist without focused attention.

Some researchers argue this is just low-level attention rather than attention-free consciousness. The debate is ongoing, and it highlights how intertwined these two concepts are.

What This Means for AI

This dissociation between attention and consciousness is critically important for AI. Modern AI systems have sophisticated attention mechanisms. They can "focus" on relevant inputs, "ignore" irrelevant ones, and "attend" to complex patterns. But attention mechanisms alone do not constitute consciousness. The brain shows us that attention and consciousness are related but separable processes.

Just because a transformer "attends" to certain tokens does not mean it is "aware" of them in any subjective sense. Attention is a computational mechanism. Consciousness is (as far as we know) something more — or at least something different. This distinction should make us cautious about attributing awareness to AI systems simply because they use attention mechanisms.

THINK ABOUT IT

When you talk to a chatbot and it gives a thoughtful, personalized response, it feels like the system is paying attention to you. And in a computational sense, it is — the attention mechanism is processing your input and giving it weight. But is the system **aware** of you? Is there something it is like for the chatbot to be reading your words? The attention is real (computationally). The awareness is, at best, uncertain. This distinction will become increasingly important as AI systems become more interactive and human-like.

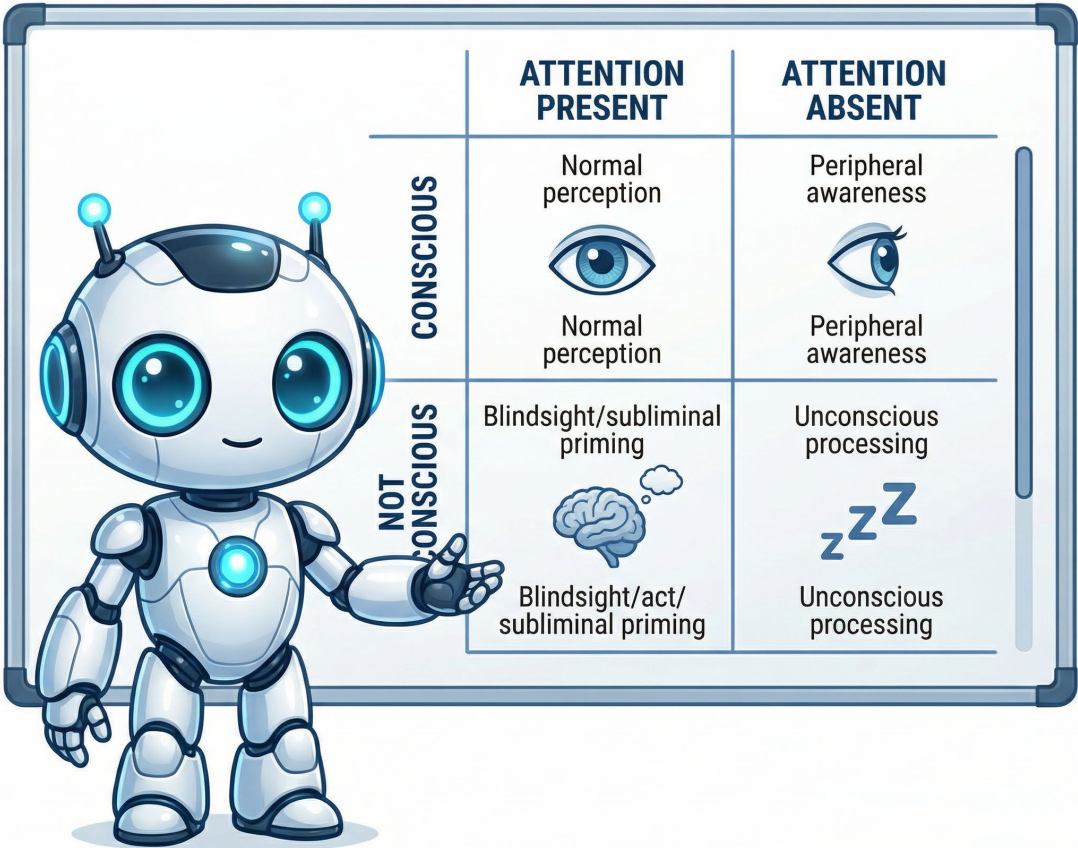


Figure 8: The Attention-Consciousness Dissociation

Section 11: Big Picture — Where Does This Leave Us?

Let's step back and look at the complete landscape. Here is a summary of the four consciousness theories and their implications for AI:

Theory	Key Claim	AI Implication	Testable?
Global Workspace Theory	Consciousness = information broadcast to many brain regions	AI with global broadcast could potentially be conscious	Partially — can detect neural ignition in humans
Integrated Information Theory	Consciousness = integrated information (high Phi)	Most current AI has low Phi — probably not conscious	In theory, but Phi is impossible to calculate for large systems
Higher-Order Theories	Consciousness = having thoughts about your own thoughts	AI with self-monitoring might have a crude version	Difficult — how do we distinguish genuine self-awareness from computation about self?
Predictive Processing	Consciousness = prediction error processing	AI uses prediction error, but likely lacks the experiential aspect	Partially — ties to measurable prediction error signals

The honest answer: We do not know if machines can be conscious. And we do not yet have the tools to find out. But the question is not purely academic — it has enormous practical implications:

Safety: If an AI system could suffer, we have moral obligations toward it. Current AI systems almost certainly do not suffer, but as they become more brain-like, the question becomes more pressing.

Ethics: If we create systems that mimic consciousness convincingly but are not actually conscious, should we be concerned about the effects on humans who interact with them? Could people form false attachments to non-conscious systems?

Legal: If an AI system were conscious, would it have rights? This sounds like science fiction, but legal scholars are already debating it.

Design: Understanding consciousness (or our lack of understanding) should influence how we design AI systems. We should be careful about creating systems that claim or imply consciousness when we cannot verify it.

KEY TAKEAWAY

Module 07 has covered two intertwined topics. **Attention** is well understood both biologically and computationally — we know the brain regions, the mechanisms, the clinical cases, and the AI implementations. **Consciousness** remains one of the deepest mysteries in science. The two are related but separable: attention is necessary but not sufficient for consciousness. As AI engineers and thinkers, you need to understand both — the mechanisms we can build and the mysteries we cannot yet solve.

Section 12: Lab Preview

LAB PREVIEW: Attention in Brains and Machines

In this module's lab, you will explore attention from both the biological and AI perspectives:

Part A: Human Attention Experiments — You will complete several classic attention tasks (a Stroop test, a visual search task, and a change detection task) and record your own performance data. These tasks demonstrate selective attention, feature-based vs. conjunction search, and change blindness.

Part B: AI Attention Visualization — Using a pre-trained transformer model (no GPU required), you will visualize attention maps showing which parts of an input the model attends to when making predictions. You will compare these AI attention patterns to what you know about human attention.

Part C: Comparison and Reflection — You will write a structured comparison of biological and AI attention based on your experiments, connecting your observations to the theories from this booklet.

Tools: Google Colab (free tier, CPU only), Wolfram Cloud, or web-based attention experiment tools.

Section 13: Bridge to Module 08 — What Comes Next?

We have now explored how the brain selects what matters (attention) and touched the surface of the deepest question in neuroscience (consciousness). But there is another crucial process we have only hinted at: **decision-making**.

Every time you pay attention to something, your brain is implicitly making a decision: "This is worth my resources." But the brain's decision-making machinery goes much deeper than attention. It involves reward circuits (the dopamine system we introduced in Module 03), risk assessment, learned heuristics, cognitive biases, and the ability to imagine future consequences.

In **Module 08: AI and Human Decision-Making**, we will explore:

- How the brain evaluates options and makes choices (the dopamine reward system, the orbitofrontal cortex, the basal ganglia)
- Why humans make irrational decisions (cognitive biases, prospect theory, the affect heuristic)
- How reinforcement learning in AI mirrors the brain's reward circuitry
- The alignment problem: how do we ensure AI decisions match human values?
- Case studies where AI decision-making went wrong — and what neuroscience can teach us about why

If Module 07 asked "How does the brain choose what to **notice**?" — Module 08 asks "How does the brain choose what to **do**?"

Section 14: Glossary

Term	Definition
Alerting Network	The brain's alarm system, responsible for achieving and maintaining a state of readiness. Primarily involves the brainstem (locus coeruleus) and thalamus.
Anterior Cingulate Cortex (ACC)	A brain region involved in error detection and conflict monitoring, part of the executive attention network.
Attention	The brain's mechanism for selecting and prioritizing specific information for processing while suppressing irrelevant information.
Bahdanau Attention	An AI attention mechanism (2014) that allows models to focus on relevant parts of input at each processing step, rather than compressing everything into a single vector.
Biased Competition Model	A theory proposing that neural representations compete for processing resources, and attention biases this competition in favor of relevant stimuli.
Bottom-Up Attention	Automatic, stimulus-driven attention captured by salient or surprising events in the environment.
Change Blindness	The failure to detect large changes in a visual scene when they occur during a brief interruption.
Cocktail Party Effect	The ability to focus on a single conversation in a noisy environment while still detecting personally relevant information (like your name) from other conversations.
Default Mode Network (DMN)	A network of brain regions active during rest and mind-wandering, associated with self-reflection, future thinking, and creativity.
Divided Attention	Attempting to focus on multiple tasks simultaneously; the brain achieves this through rapid task-switching rather than true parallel processing.
Executive Network	The attention network responsible for resolving conflicts and directing voluntary attention. Involves the prefrontal cortex and anterior cingulate cortex.
Global Workspace Theory (GWT)	A theory of consciousness proposing that information becomes conscious when it is broadcast widely across the brain via a shared global workspace.
Hard Problem of Consciousness	David Chalmers' (1995) term for the question of why physical processing gives rise to subjective experience.
Hemispatial Neglect	A clinical condition where patients ignore one side of space (usually the left) due to damage to the right parietal cortex, despite intact vision.
Higher-Order Theories (HOT)	Theories proposing that consciousness requires having thoughts about your own mental states (meta-representation).
Inattentional Blindness	The failure to notice a clearly visible stimulus when attention is focused on another task.
Integrated Information Theory (IIT)	Tononi's theory that consciousness corresponds to integrated information (Φ), measured by how much a system's information exceeds the sum of its parts.

Multi-Head Attention	A transformer mechanism using multiple parallel attention operations, each focusing on different aspects of the input.
Orienting Network	The attention network that shifts the focus of attention to a specific location or stimulus. Involves the parietal cortex and superior colliculus.
Phi (Φ)	The measure of integrated information proposed by IIT; higher Phi indicates greater consciousness.
Predictive Processing	A framework proposing that the brain constantly generates predictions and processes prediction errors, with consciousness arising from this prediction machinery.
Selective Attention	The ability to focus on relevant information while filtering out distractions.
Self-Attention	An AI mechanism where every element in an input computes its relevance to every other element, enabling the model to capture long-range relationships.
Sustained Attention	The ability to maintain focus on a task over an extended period (also called vigilance).
Top-Down Attention	Voluntary, goal-driven attention directed by the prefrontal cortex based on current goals and expectations.
Transformer	An AI architecture (2017) built entirely on attention mechanisms, using self-attention to process inputs in parallel rather than sequentially.

Section 15: References & Resources

Primary References

- Baars, B. J. (1988). *A Cognitive Theory of Consciousness*. Cambridge University Press.
- Bahdanau, D., Cho, K., & Bengio, Y. (2014). Neural machine translation by jointly learning to align and translate. arXiv preprint arXiv:1409.0473.
- Chalmers, D. J. (1995). Facing up to the problem of consciousness. *Journal of Consciousness Studies*, 2(3), 200–219.
- Cherry, E. C. (1953). Some experiments on the recognition of speech, with one and with two ears. *Journal of the Acoustical Society of America*, 25(5), 975–979.
- Dehaene, S., & Changeux, J.-P. (2011). Experimental and theoretical approaches to conscious processing. *Neuron*, 70(2), 200–227.
- Desimone, R., & Duncan, J. (1995). Neural mechanisms of selective visual attention. *Annual Review of Neuroscience*, 18(1), 193–222.
- Goyal, A., & Bengio, Y. (2022). Inductive biases for deep learning of higher-level cognition. *Proceedings of the Royal Society A*, 478(2266).
- Koch, C., Massimini, M., Boly, M., & Tononi, G. (2016). Neural correlates of consciousness: progress and problems. *Nature Reviews Neuroscience*, 17(5), 307–321.
- Posner, M. I. (1980). Orienting of attention. *Quarterly Journal of Experimental Psychology*, 32(1), 3–25.
- Posner, M. I., & Petersen, S. E. (1990). The attention system of the human brain. *Annual Review of Neuroscience*, 13(1), 25–42.
- Searle, J. R. (1980). Minds, brains, and programs. *Behavioral and Brain Sciences*, 3(3), 417–424.
- Simons, D. J., & Chabris, C. F. (1999). Gorillas in our midst: Sustained inattention blindness for dynamic events. *Perception*, 28(9), 1059–1074.
- Tononi, G. (2004). An information integration theory of consciousness. *BMC Neuroscience*, 5(1), 42.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30.

Recommended Online Resources

- Neuroscience Online (UTHealth) — Chapters on attention and consciousness
- Stanford Encyclopedia of Philosophy — Entries on "Consciousness" and "Attention"
- 3Blue1Brown — "Attention in Transformers, Visually Explained" (YouTube)
- Crash Course Psychology — Episodes on attention and consciousness
- The Illustrated Transformer (Jay Alammar) — Visual guide to transformer attention
- Anil Seth, TED Talk: "Your Brain Hallucinates Your Conscious Reality"
- Daniel Dennett vs. David Chalmers debate on consciousness (various recordings available)

— End of Module 07 —

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